IT2011 - Artificial Intelligence and Machine Learning

Department of Information Technology, Faculty of Computing

Year 2 semester 1 (2026)

Tutorial 07

Unsupervised Learning Workshop 1

K-Means Clustering

Question 01

Consider the following dataset of two-dimensional points:

Point	X_1	X_2
P	1	4
Q	2	5
R	3	3
S	8	7
T	9	8
U	8	9

Assume that the number of clusters K = 2, and the initial centroids are selected as: Centroid C1 = (1, 4) and Centroid C2 = (8, 7).

i. The Euclidean distance between two points $(x_1, y_1)$and $(x_2, y_2)$is defined as:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Using this formula, compute the distance from each data point in the dataset to both initial centroids C_1 and C_2 . Clearly show all intermediate calculation steps and summarize your results in a tabular format indicating distances to each centroid.

ii. Based on the distance values calculated in part (b), assign each data point to the nearest centroid. Clearly justify your assignment and list the points belonging to each cluster.

iii. After forming the clusters, compute the new centroids by calculating the mean of the coordinates of all points within each cluster.

iv. Using the newly calculated centroids, perform one additional iteration of the K-Means algorithm. Recalculate the distances, reassign the clusters, and determine whether the algorithm has converged. Provide justification for your answer.

v. After performing K-Means clustering for different values of K, the following Within-Cluster Sum of Squares (WCSS) values are obtained:

K	2	3	4	5	6	7	8
WCSS	300	180	110	85	80	78	77

vi. Using the Elbow Method, analyze the above values and determine the most appropriate number of clusters.
Provide a clear justification for your answer based on the observed pattern.

Question 02

You are given a dataset clustered into two groups:

Point	X	Y	Cluster
A	1	2	C1
B	2	3	C1
C	3	2	C1
D	8	7	C2
E	9	8	C2
F	8	9	C2

i. The Euclidean distance between two points $(x_1, y_1)$and $(x_2, y_2)$is defined as:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Using this formula,

a. calculate the distances between all points in Cluster 1 (Intra Cluster Distance within C1). Clearly show all intermediate calculation steps.

b. calculate the distances between all points in Cluster 2 (Intra Cluster Distance within C2). Clearly show all intermediate calculation steps.
c. calculate the distance between every point in Cluster 1 and every point in Cluster 2 (Inter Cluster Distance). Clearly show all intermediate calculation steps and summarize your results in a tabular format.

ii. Using the formula,

$$Dunn\ Index = \frac{\min\ (Inter\ cluster\ Distance)}{\max\ (Intra\ cluster\ Distance)}$$

a. find the minimum Inter Cluster Distance.
b. find the maximum Intra Cluster Distance.
c. calculate the Dunn Index.

iii. What does the value of the Dunn Index indicate about the quality of clustering? Based on your result, is this clustering good or poor? Justify your answer.